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- 📊 Results
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🏃 Human Activity Recognition

24AI636 · Mini Project 4 · CNN + LSTM Hybrid (DeepConvLSTM) · UCI HAR Dataset

📋 Project Overview

Problem Definition

Human Activity Recognition (HAR) using smartphone inertial sensors is a core challenge in **ubiquitous computing** and **health monitoring**.

- **Industry relevance**: Fitness apps, elder care, rehabilitation monitoring
- **Dataset**: UCI HAR Dataset — 10,299 samples, 9 sensor signals, 128 timesteps
- **Classes**: 6 activities (Walking, Upstairs, Downstairs, Sitting, Standing, Laying)

Why CNN + LSTM?

Aspect	Reason
CNN	Extracts local motion patterns from raw sensor windows
LSTM	Learns temporal dynamics and activity transitions
Combined	Spatiotemporal feature representation → best accuracy

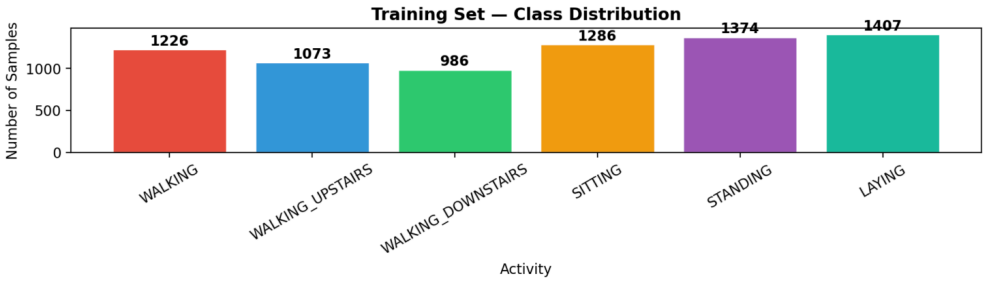
📦 Dataset Split

Split	Samples	Sub
Train	7352	
Test	2947	

🔧 Data Engineering

- ✅ Window normalization (z-score)
- ✅ Gaussian noise augmentation (+30%)
- ✅ Class balance verification
- ✅ 9 sensor channels retained

📊 Activity Class Distribution



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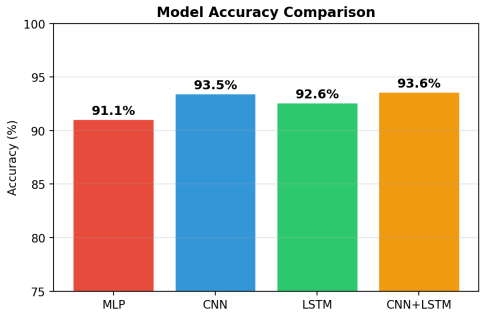
📊 Experimental Results

ACCURACY	PRECISION	RECALL	F1-SCORE
93.6%	93.6%	93.6%	93.6%

🔬 Model Comparison



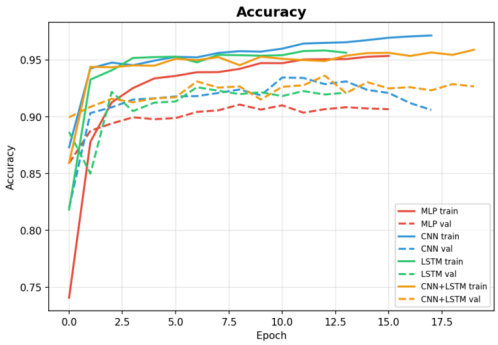
📋 Detailed Metrics

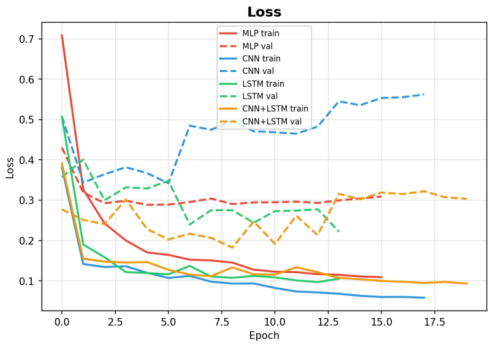
Model	Accuracy	Precision	Recall
MLP	91.08%	91.18%	91.08%
CNN	93.45%	93.46%	93.45%
LSTM	92.6%	92.71%	92.6%
CNN+LSTM	93.62%	93.63%	93.62%

🔧 Best Hyperparameters

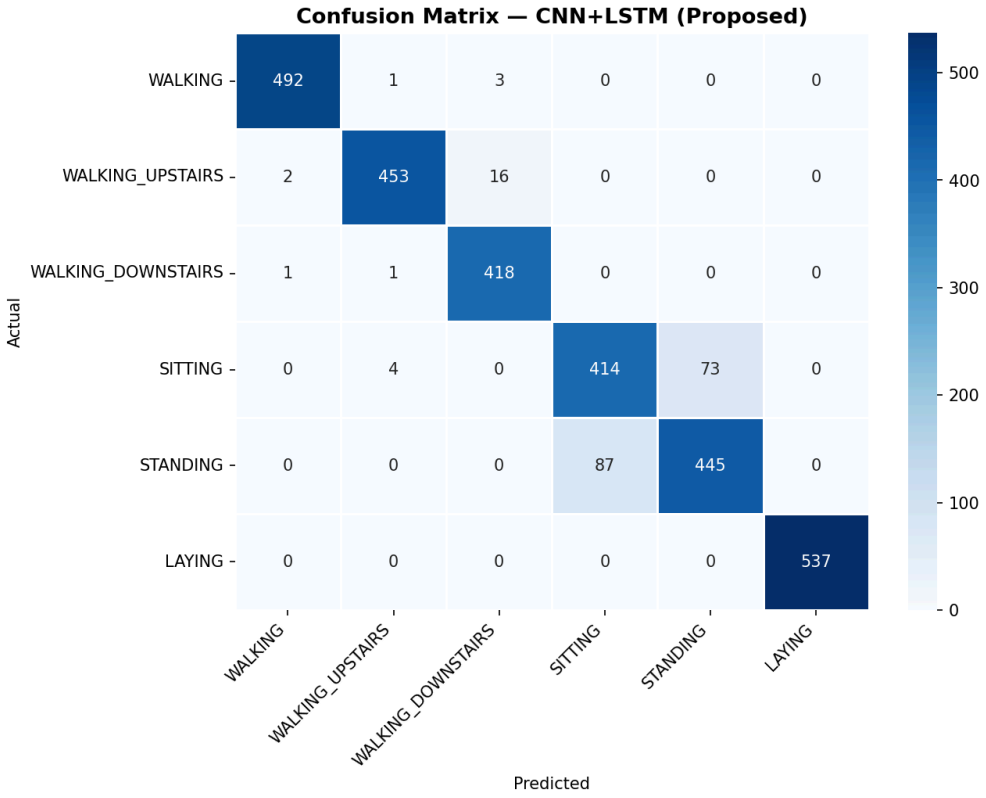
Parameter	Value
CNN Filters	64 → 128
Kernel Size	3
LSTM Units	100
Dropout	0.3
Learning Rate	0.001
Batch Size	64

📈 Training Curves





🎯 Confusion Matrix — CNN+LSTM



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🔮 Live Activity Prediction

Simulate real-time sensor input and watch the model classify your activity.

⚙️ Sensor Input Simulation

Simulate Activity

SITTING

Sensor Noise Level

0.35

Prediction Samples

11

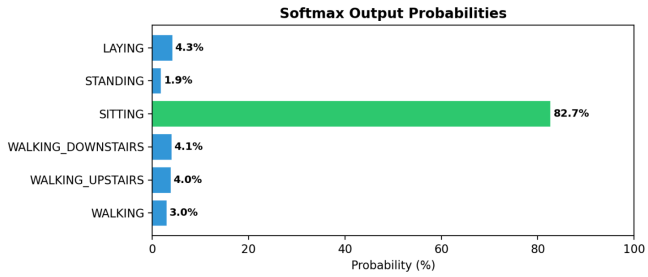
🏃 Run Prediction

Predicted Activity

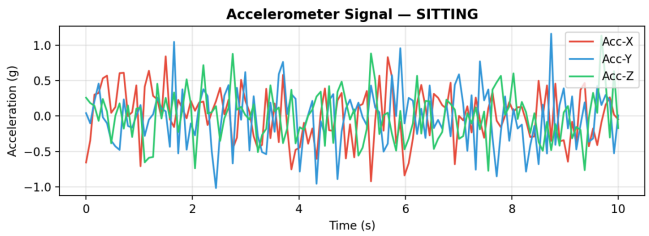
🪑 SITTING

✅ CORRECT — Confidence: 82.7%

📊 Class Probability Distribution



📡 Simulated Sensor Signal (Sample 1)



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📐 Model Architecture — DeepConvLSTM

🧠 Architecture Justification

HAR signals have **two key properties**:

1. **Spatial (local)**: Short bursts of motion repeated in a window
2. **Temporal (sequence)**: Activities unfold over time as state transitions

CNN → extracts local feature maps from sensor windows
LSTM → models temporal dependencies between extracted features

🔥 Architecture Diagram

🏗️ Layer-by-Layer

Layer	Output Shape	Parameters
Input	(128, 9)	—
Conv1D (64 filters, k=3)	(128, 64)	1,792
BatchNorm + MaxPool	(64, 64)	256
Conv1D (128 filters, k=3)	(64, 128)	24,704
BatchNorm + MaxPool	(32, 128)	512
LSTM (100 units)	(100,)	91,600
Dropout (0.3)	(100,)	—
Dense (64)	(64,)	6,464
Softmax (6)	(6,)	390
Total	—	**~125K params**

🔮 Viva Answer

> *"HAR data has both temporal and spatial dependencies.*
> *CNN captures local sensor patterns from the window, while LSTM captures temporal dynamics over time.*